PH.40121 **FALL 2026**

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 7: Conservation Laws and the Newtonian Limit

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Overview

The Jacobian of the reference observables is used to enforce both the Newtonian limit and conservation constraints.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$J_{ai} = \partial R_a / \partial \theta_i$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$\alpha_1 = -1/2$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$\alpha_2 = -1/4$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

$$J \approx \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$$

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Einstein–Mythos paper C, §§4–5
- SGT9 dependency graph, sheet 3
- Midterm Review Manual